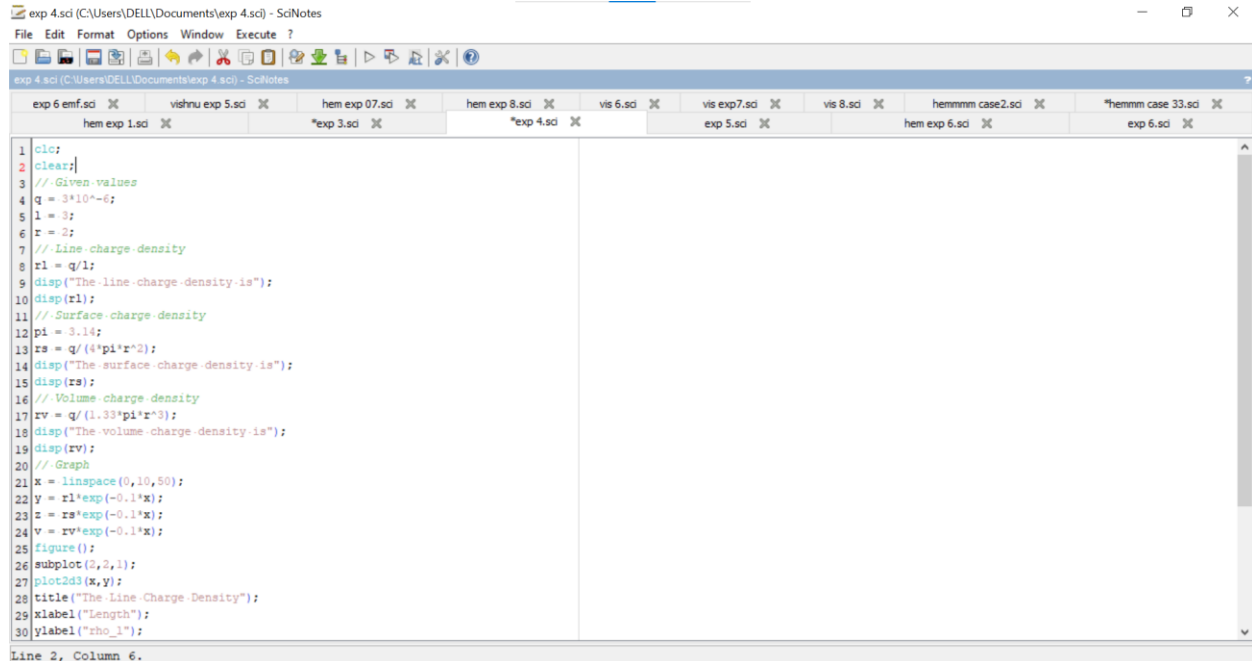


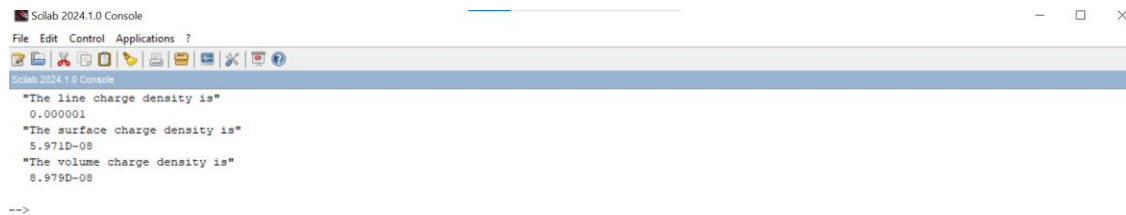
EXP 4: Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.



The screenshot shows the SciNotes application with a script for calculating charge densities. The script is as follows:

```
1 clc;
2 clear;
3 // Given values
4 q = 3*10^-6;
5 l = 3;
6 r = 2;
7 // Line charge density
8 rl = q/l;
9 disp("The line charge density is");
10 disp(rl);
11 // Surface charge density
12 pi = 3.14;
13 rs = q/(4*pi*r^2);
14 disp("The surface charge density is");
15 disp(rs);
16 // Volume charge density
17 rv = q/(1.33*pi*r^3);
18 disp("The volume charge density is");
19 disp(rv);
20 // Graph
21 x = linspace(0,10,50);
22 y = rl*exp(-0.1*x);
23 z = rs*exp(-0.1*x);
24 v = rv*exp(-0.1*x);
25 figure();
26 subplot(2,2,1);
27 plot2d3(x,y);
28 title("The Line Charge Density");
29 xlabel("Length");
30 ylabel("rho_l");
```

Line 2, Column 6.



The screenshot shows the SciLab 2024.1.0 Console with the following output:

```
"The line charge density is"
0.000001
"The surface charge density is"
5.971D-08
"The volume charge density is"
8.979D-08
```

-->

Exp 4:-

charge value = 3×10^{-6}

length value = 3 mm

Radius value = 2 cm

Sol

$$P_L = \frac{3 \times 10^{-6}}{3 \times 10^{-3}}$$

$$P_L = 1 \times 10^{-3} \text{ C/m}$$

$$P_L = 0.001 \text{ C/m}$$

$$R = 2 \times 10^{-2} \text{ m}$$

$$P_R = \frac{3 \times 10^{-6}}{4\pi (2 \times 10^{-2})^2}$$

$$= 3 \times 10^{-6}$$

$$\rho_v = \frac{3 \times 10^{-6}}{3.51 \times 10^{-5}}$$

$$\rho_v = 8.95 \times 10^{-3} \text{ c/m}^3$$

o/p ✓

dm ✓